

Question Paper

Exam Date & Time: 11-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING / INFORMATION TECHNOLOGY) DEGREE
EXAMINATIONS - JUNE 2025
SUBJECT: MAT 2226/MAT_2226/MAT_2256 - ENGINEERING MATHEMATICS - IV

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) In n tosses of a fair coin, let A be the event that there is at least one head and one tail. Let B be the event that there is at most one head. What is the value of n such that A and B^c are independent? (4)

- 1B) In a certain town 40% have brown hair, 25% have brown eyes, 15% have both brown hair and brown eyes. A person is selected at random. (3)
- i) If he has brown hair, then what is the probability that he has brown eyes also.
 - ii) If he has brown eyes, then what is the probability that he does not have brown hair.
 - iii) Determine the probability that he has neither brown hair nor brown eyes.

- 1C) The amount X of bread in hundreds of kilos that a bakery sells in a day is a random variable with probability density function (3)

$$f(x) = \begin{cases} cx, & 0 \leq x \leq 3 \\ c(6-x), & 3 \leq x \leq 6 \\ 0, & \text{elsewhere} \end{cases}$$

- i. Find the value of c which makes $f(x)$ a pdf.
- ii. What is the probability that the amount of bread (in hundreds of kilos) that will be sold in a day is between 1.5 and 2.5.

- 2A) In an intelligence test administered to 1000 children, the average score is 42 and standard deviation 24. (4)
- (Assume the normal distribution)
- i) Find the number of children exceeding the score 60.
 - ii) Find the number of children with score lying between 20 and 40.

- 2B) The probability of a man hitting a target is $1/3$. How many times must he try so that the probability of hitting the target at least once is more than 90%? (3)

- 2C) For random variables (X, Y) with joint pdf $f(x, y) = 8xy$, $0 < x < y < 1$, compute the coefficient of correlation ρ . (3)

- 3A) Let X has binomial distribution with $n = 10$,
 $p \in \left[p: p = \frac{1}{4}, \frac{1}{2} \right]$. To test the simple
hypothesis $H_0: p = \frac{1}{2}$ against the alternative
hypothesis $H_1: p = \frac{1}{4}$ a random sample X_1 is
used with the critical region $C =$
 $\{x_1 \mid x_1 \leq 3\}$. Then find the power function of
the test and hence find the significance level. (4)

- 3B) In a factory Machines A, B, C are all producing chips of same length. Machines A, B, C produce
1%, 4%, 2% defective chips respectively. Of the total production of chips in the factory, Machine A
produces 30%, B produces 25%, and C produces 45%. (3)
i) If one chip is selected at random from the total chips produced in a given day, determine the
probability that it is defective.
ii) Given that the selected chip is defective, find the probability that it was produced by Machine B.

- 3C) Let $(X_1, X_2, \dots, X_n)$ be a random sample of size n
from a distribution whose pmf is given by (3)

$$f(x; \theta) = \begin{cases} \theta^x (1 - \theta)^{1-x} & x = 0, 1 \text{ and } 0 \leq \theta \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the maximum likelihood estimator for θ .

- 4A) Find $P(X \leq 1 \mid Y \leq 3)$, $V(X)$ and $V(Y)$
for the following distribution of (X, Y) . (4)

$X \backslash Y$	1	2	3	4	5	6
0	0	0	1/32	2/32	2/32	3/32
1	1/16	1/16	1/8	1/8	1/8	1/8
2	1/32	1/32	1/64	1/64	0	2/64

- 4B) Let X and Y be independent random variables such
that $X \sim N(4, 25)$ and $Y \sim N(3, 16)$. Use moment
generating functions to show that $Z = 5X - 4Y + 7$
has a normal distribution. Find the mean and
variance of Z . (3)

- 4C) The time for which a machine in a factory can run without having to be restarted is a random
variable having the exponential distribution with mean 30 days. Find the probabilities that (3)
i) the machine will have to be restarted in less than 10 days.
ii) the machine will not have to be restarted in at least 45 days.

- 5A) A survey of 319 families with 5 children each reveals the following distribution (4)

Number of boys	5	4	3	2	1	0
Number of families	14	55	110	88	40	12

Is this result consistent with the hypothesis that the male and female birth are equally probable at
0.05 significance level.

- 5B) (3)

An economist wishes to estimate the average family income in a certain population. The population standard deviation is known to be 4,500 dollars and the economist uses the random sample of size $n = 225$. What is the probability that the sample mean will fall within 800 dollars of the population mean?

- 5C) Let X be uniformly distributed over $[0, 1]$ and $Y = \frac{1}{X+1}$. Then find the pdf of Y . (3)

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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025		MAT-2256:Engineering Mathematics - IV	
#	Course Outcome	Description	Marks

No records found / CO Level Description not defined

* **NA** - CO Not Selected

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